

Pre-Training of Large Language Models

Part 1 : Task and loss

Language Models(LM) :

A simple probability like function which tokenize the sentence and use the first and n words to make out the relationship and give us the main probability

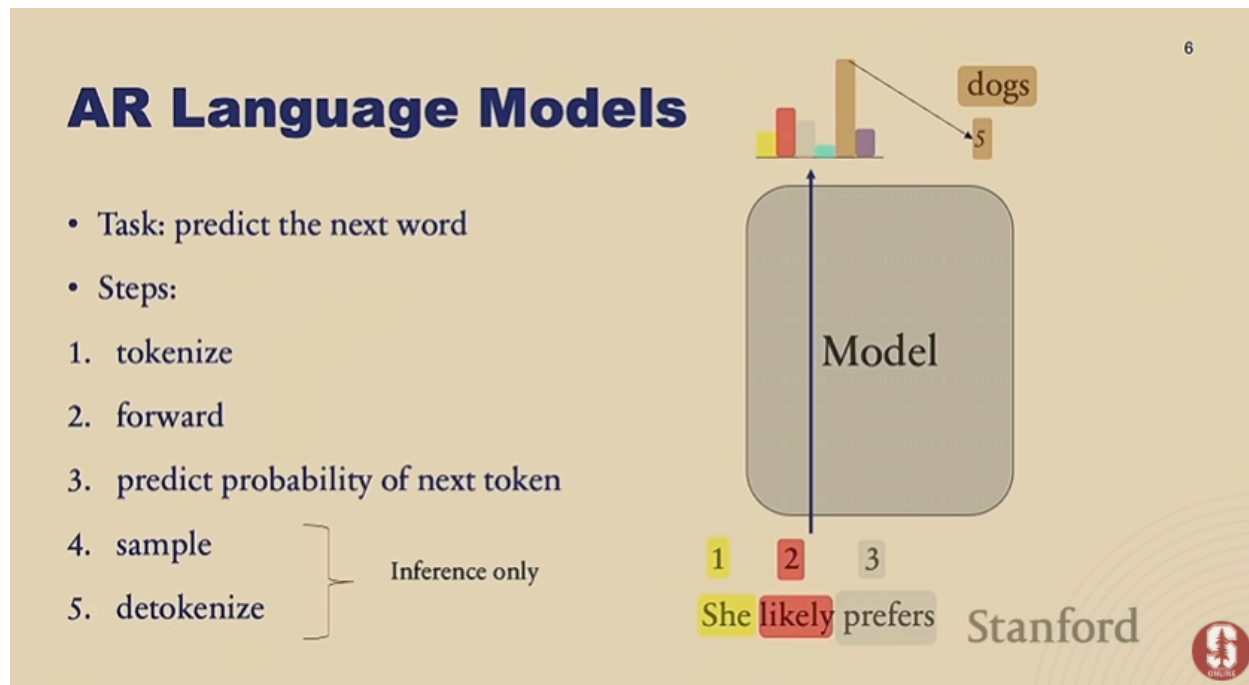
Autoregressive models(AR) :

Auto means "self."

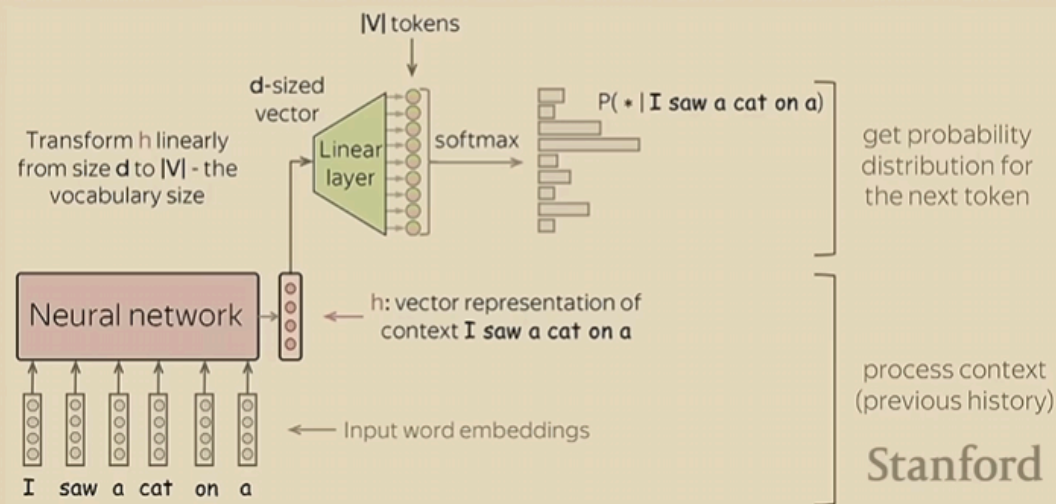
Regression is a statistical method used to predict relationships between variables.

Put together, an AR model predicts a variable's future by looking exclusively at that same variable's history. It operates on the core assumption that past patterns have a direct, mathematical influence on what happens next.

Where it is used: You will see AR models used frequently in finance to predict stock prices, in economics to forecast inflation, and in business to predict future sales volume based on last month's numbers.



AR Neural Language Models

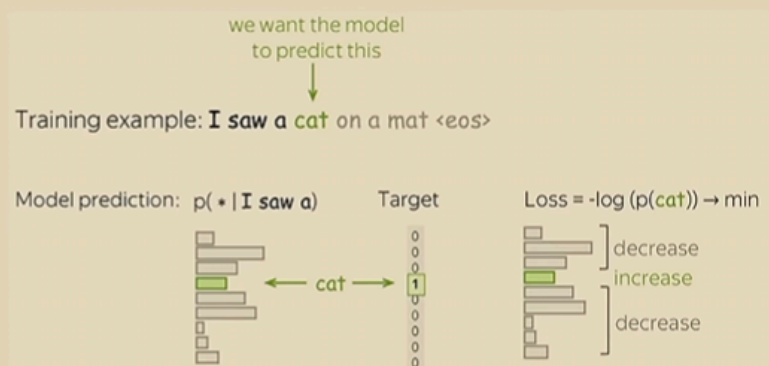


For calculating the loss :

Loss

- Classify next tokens' index

- $\Rightarrow$ cross-entropy loss



- $\Rightarrow$ maximize text's log-likelihood

$$\max \prod_i p(x_i | x_{1:i-1}) = \min \left(- \sum_i \log p(x_i | x_{1:i-1}) \right) = \min \mathcal{L}(x_{1:L})$$

Tokenizer

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Tokenizer

- Why?
 - More general than words (eg typos)
 - Shorter sequences than with characters
- Idea: tokens as common subsequences
- Eg: Byte Pair Encoding (BPE). Train steps:
 1. Take large corpus of text
 2. Start with one token per character
 3. Merge common pairs of tokens into a token
 4. Repeat until desired vocab size or all merged

tokenizer:
text to token
index

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tokenizer:  
text to token index
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Part 2 : Evaluation

LLM Evaluation perplexity :

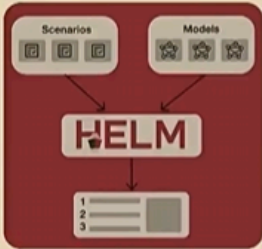
If an LLM has a perplexity of 10 on a given text, it means that, on average, the model was as uncertain as if it were rolling a 10-sided die to choose the next word from 10 equally likely options.

- Perplexity of 1: The model is a perfect oracle; it predicts the exact next word with 100% confidence every time.
- Low Perplexity (e.g., 10-20): The model has a strong grasp of the language patterns and is confident in its predictions.
- High Perplexity (e.g., 100+): The model is confused, often indicating that the input text is ambiguous, outside its training data, or structurally broken

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LLM Evaluation: agg. std NLP benchmarks

Holistic evaluation of language models (HELM)



Model	Mean ratio
GPT-4 (0613)	0.902
GPT-4 Turbo (1106 preview)	0.834
Palmira X V3 (72B)	0.821
Palmira X V2 (33B)	0.783
PaLM-2 (Unicom)	0.776
Yi (34B)	0.772

SEE MORE

Huggingface open LLM leaderboard



collect many automatically evaluable benchmarks, evaluate across them

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Why We Use It :

Computationally Cheap: Unlike human evaluations or "LLM-as-a-judge" methods, perplexity is calculated automatically during the model's standard forward pass, making it incredibly fast to compute.

Real-Time Feedback: It is heavily used during the actual training phase of an LLM to monitor if the model is successfully learning to minimize its error rate.

Standard Benchmark: It has been the historical standard for evaluating statistical language models since the 1970s.

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LLM Evaluation: agg. std NLP benchmarks

- Mix of things that can be “easily” evaluated
- Typically there is “gold” answer
=> you likelihood of LLM to predict that vs other options

Scenario	Task	What	Who
NarrativeQA narrative_qa	short-answer question answering	passages are books and movie scripts, questions are unknown	annotators from summaries
NaturalQuestions (closed-book) natural_qa_closedbook	short-answer question answering	passages from Wikipedia, questions from search queries	web users
NaturalQuestions (open-book) natural_qa_openbook_fergans	short-answer question answering	passages from Wikipedia, questions from search queries	web users
OpenbookQA openbookqa	multiple-choice question answering	elementary science	Amazon Mechanical Turk workers
MMLU (Massive Multitask Language Understanding) mmlu	multiple-choice question answering	math, science, history, etc.	various online sources
GSM8K (Grade School Math) gsm	numeric answer question answering	grade school math word problems	contractors on Upwork and Surge AI
MATH math_chain_of_thought	numeric answer question answering	math competitions (AMC, AIME, etc.)	problem solvers
LegalBench legalbench	multiple-choice question answering	public legal and administrative documents, manually constructed questions	lawyers
MedQA med_qa	multiple-choice question answering	US medical licensing exams	problem solvers
WMT 2014 wmt_14	machine translation	multilingual sentences	European, news, Common Crawl

HELM-lite
[Liang+ 2022]

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ONLINE

LLM Evaluation: eg MMLU

- Example: MMLU
- ~Most trusted pretraining benchmark

Astronomy

What is true for a type-Ia supernova?

- A. This type occurs in binary systems.
- B. This type occurs in young galaxies.
- C. This type produces gamma-ray bursts.
- D. This type produces high amounts of X-rays.

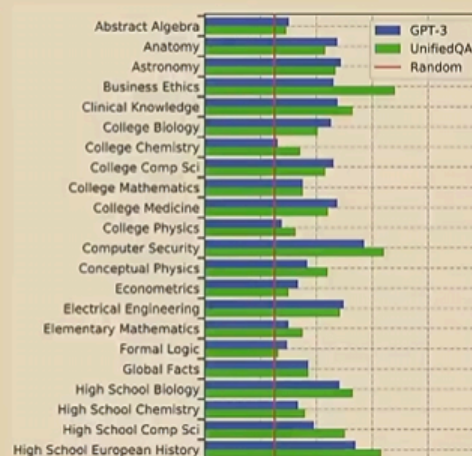
Answer: A

High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

- A. directional selection.
- B. stabilizing selection.
- C. sexual selection.
- D. disruptive selection

Answer: A



MMLU
[Hendrycks+ 2020]



Limitations of Perplexity :

It Does Not Measure Truth or Understanding: A model might string together perfectly coherent, grammatically flawless words (resulting in low perplexity) that are factually incorrect or completely nonsensical.

Tokenization Dependency: You cannot compare the perplexity scores of two different models if they use different tokenizers (e.g., comparing Llama 3 to GPT-4). Because the models chop words into differently sized pieces, the math relies on different denominators, rendering direct comparisons meaningless.

When someone builds a new AI, how do we know if it is smarter than the previous one?

Instead of sitting and chatting with it for hours, researchers collect dozens of existing multiple-choice exams, bundle them together (aggregate them), and run the model through the test automatically.

- **HELM (Holistic Evaluation of Language Models):** A massive benchmark suite built by Stanford. Instead of testing just one skill, it grades models on accuracy, bias, toxicity, and reasoning across many scenarios.
- **Hugging Face Open LLM Leaderboard:** A public scoreboard where open-source models are ranked based on their average test scores, showing who currently holds the top spot.

2. How Automated Testing Works

To test a model automatically without human graders, researchers need tests with a "Gold" answer meaning there is one undisputed correct answer key.

Benchmark	What It Tests	Example Question
MMLU	General knowledge across 57 academic subjects	Multiple-choice questions from elementary school to college level.
GSM8k	Grade School Math (8,000 word problems)	<i>"Jason had 20 lollipops. He gave half to his sister and ate 3. How many are left?"</i>
MATH	Advanced high school & olympiad math	Multi-step calculus, algebra, and geometry proofs.
MedQA	Medical knowledge	Real questions taken from US medical board licensing exams.
LegalBench	Legal reasoning	Analyzing clauses, legal precedents, and contract terms.

Because the correct answer is fixed (e.g., Option A or the number 7), a script calculates the percentage of correct answers in seconds.

Why Researchers Use Them (and the Catch)

- Why they are useful: They are fast, objective, cheap to run, and allow direct head-to-head comparisons across different models.
- The Catch ("Data Contamination"): Because all these test questions are publicly available on the internet, models can accidentally memorize the questions during training—like a student who stole the exam answer key before the test. This has led researchers to continuously build harder benchmarks (like MMLU-Pro and GPQA).

Part 3 : Data

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Data

- Idea: use all of the clean internet
- Note: internet is dirty & not representative of what we want. Practice:
 1. Download all of internet. Common crawl: 250 billion pages, > 1PB (>1e6 GB)
 2. Text extraction from HTML (challenges: math, boiler plate)
 3. Filter undesirable content (e.g. NSFW, harmful content, PII)
 4. Deduplicates (url/document/line). E.g. all the headers/footers/menu in forums are always same
 5. Heuristic filtering. Rm low quality documents (e.g. # words, word length, outlier toks, dirty toks)
 6. Model based filtering. Predict if page could be references by Wikipedia.
 7. Data mix. Classify data categories (code/books/entertainment). Reweight domains using scaling laws to get high downstream performance.

We need to crawl data from google and you can see in the image how many pages are there and still getting increased day by day and then preprocess the data by applying the steps given in the image

Data

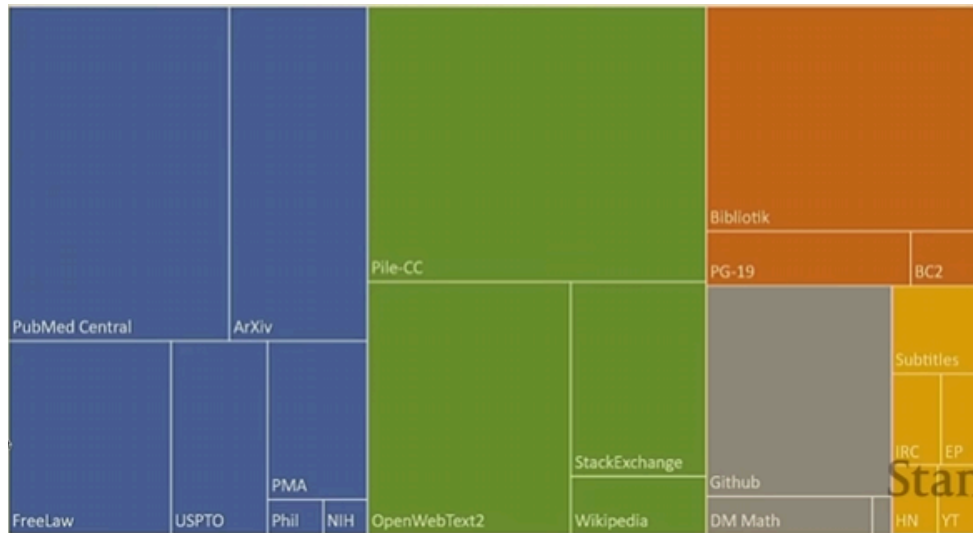
- Collecting well data is a **huge** part of practical LLM (~the key)
- Lot of research to be done!
 - How do you process well and efficiently?
 - Synthetic data?
 - How do you balance domains?
 - Multi-modal data?
- A lot of secrecy:
 - Competitive dynamics
 - Copyright liability
- Common academic datasets:
 - C4 (150B tokens | 800GB)
 - Dolma (3T tokens)
 - The Pile (280B tokens)
 - FineWeb (15T tokens)
- Closed: LLaMA 2 (2T tokens), LLaMA 3 (15T tokens), GPT-4 (~13T tokens?)

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The Importance and Secrecy of Data

- Data is the Secret Sauce: Collecting good data is a huge part, and practically "the key," to building a successful practical LLM.
- Unsolved Challenges: Researchers are actively studying how to efficiently process this information, balance different domains of knowledge, and incorporate "synthetic" (AI-generated) or "multi-modal" (images and audio) data.
- Hidden Recipes: Tech companies keep their exact data sources highly secret due to "competitive dynamics" (staying ahead of rivals) and "copyright liability" (avoiding lawsuits over using protected books or articles).
- Mind-Boggling Scale: Data is measured in "tokens" (pieces of words). Common open academic datasets range from C4 at 150 Billion tokens (which takes up 800GB of space) up to FineWeb at 15 Trillion tokens. Closed, commercial models use similarly massive amounts, like LLaMA 3's 15 Trillion tokens and GPT-4's estimated 13 Trillion tokens.



What is Actually Inside the Data?

This image provides a visual map of a specific famous dataset called "The Pile," showing exactly what categories of text make up its massive size. To make an AI smart, you cannot just feed it random websites; it needs a balanced educational diet:

- Academic (Blue section): Contains dense, professional materials like medical research (PubMed Central), science papers (ArXiv), and legal or patent documents (FreeLaw, USPTO). This gives the AI its high-level expertise.
- Internet (Green section): Includes general web scrapes (Pile-CC, OpenWebText2), encyclopedias (Wikipedia), and Q&A forums (StackExchange). This provides broad factual knowledge.
- Prose (Orange section): Made up of books and literature (such as Bibliotik and PG-19). This teaches AI grammar, long-form writing, and storytelling.
- Dialogue (Yellow section): Includes movie subtitles and chat logs (like IRC and YT). This teaches the AI how to converse naturally back and forth.
- Misc (Gray section): Includes raw computer code repositories (Github) and math datasets (DM Math) to teach the AI logic and programming.

Common Terminologies

Tokens vs. Parameters

- **Tokens (The Study Material):** Tokens are the fundamental "building blocks" or "units of thought" of the data itself. You can think of them as pieces of words. When a company says they trained an AI on 12 trillion tokens, they are talking about the massive volume of text (books, articles, websites) the AI was forced to read during its training phase.
- **Parameters (The Brain Size):** Parameters are the internal variables, weights, and rules inside the AI model. If you think of an AI model like a video game, tokens are the inputs and actions you provide, while parameters are the game's underlying engine settings that control exactly how the game reacts to your inputs. A 7B parameter model (7 billion parameters) refers to the size, complexity, and capability of that internal engine.

In short: Tokens are the *inputs* the AI reads, and parameters are the *internal rules and knowledge* it forms after processing that data.

What is the C4 Dataset?

C4 stands for the Colossal Clean Crawled Corpus. It is a massive dataset of English text developed by Google engineers to train AI models. They created it by taking a snapshot of "Common Crawl"—which is essentially a raw, unfiltered download of the public internet—and applying strict filters to it. By cleaning out gibberish, duplicate pages, and bad formatting, they created a massive, high-quality dataset used to teach language models how to write naturally.

Commercial Models vs. Open Models

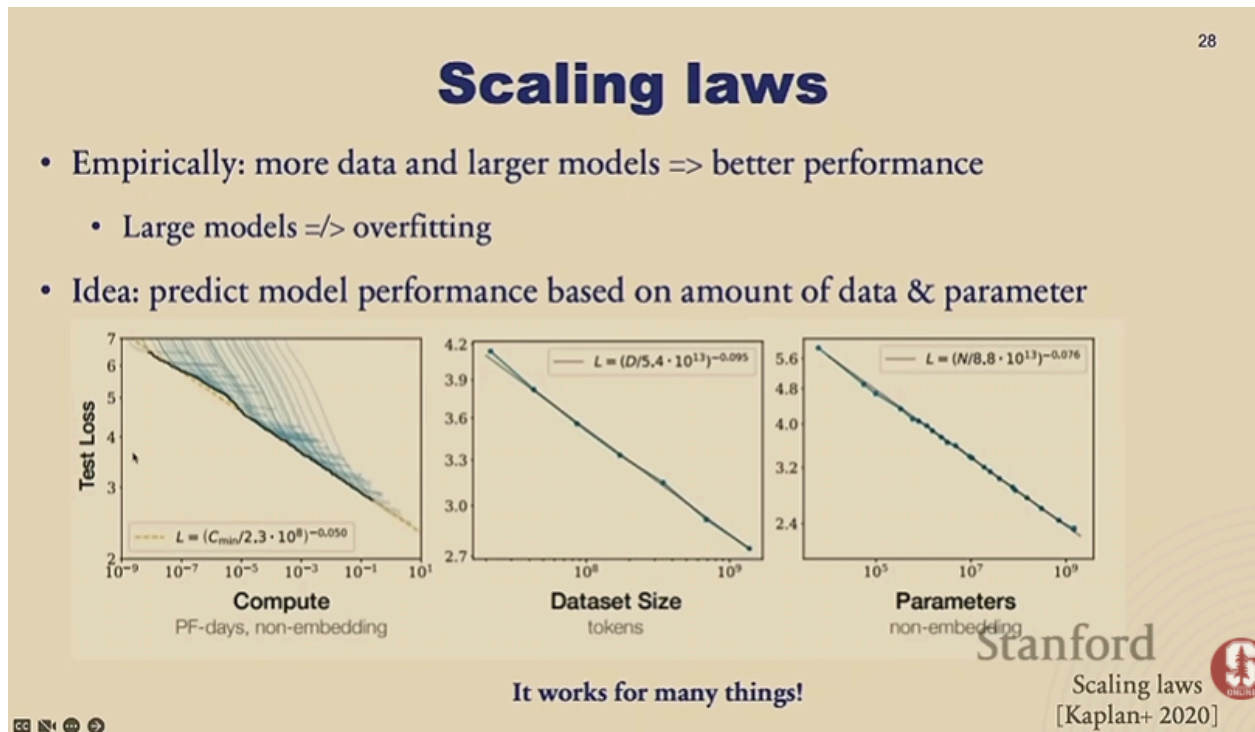
- **Open Models (Open-Weights):** These are models like Meta's Llama 3 or Google's Gemma. The creators allow anyone to freely download the actual model (the 7 billion parameters) to their own local machine. As a developer, you can run them offline, hook them directly into your Python backends or React interfaces, and fine-tune their behavior without paying any API fees.
- **Commercial Models (Closed):** These are models like OpenAI's GPT-4 or Anthropic's Claude 3. The companies keep the parameters securely locked away on their own private servers to protect their intellectual property. You cannot download the model itself; you can only interact with it by sending tokens back and forth over the internet via an API or a paid web dashboard.

Part 4 : Scaling Laws

Scaling Laws (discovered by OpenAI researchers led by Jared Kaplan in 2020).

Before this paper, training giant AI models felt like gambling. Spending \$50 million on GPUs was a huge risk because nobody was certain whether making a model 10 times bigger would actually make it smarter.

Scaling laws proved that AI performance is predictable like physics.



1. The Core Idea

- Empirical Observation: When you increase both the model size (parameters) and the amount of reading material (tokens), the model reliably gets better.
- Predictability: Instead of guessing, researchers can use a mathematical formula (a power law) to calculate in advance: *"If I give this model X amount of compute and Y amount of data, its error rate will drop by exactly Z."*

2. Reading the Three Graphs

On all three graphs, the vertical axis (Y-axis) is Test Loss.

- Loss is simply the model's error rate. Lower loss means better predictions, higher accuracy, and lower perplexity.
- Notice that all three graphs form straight lines sloping downward. Because these are plotted on a logarithmic scale, a straight line means a predictable power-law relationship:
 1. Compute (Left Graph): As you throw more total computational work at the training run (measured in GPU days), the error steadily drops.
 2. Dataset Size (Middle Graph): As you feed the model more tokens (from millions to billions to trillions), the error drops along a straight line.
 3. Parameters (Right Graph): As you increase the number of parameters (from 10M to 100M to billions of weights), the error continues to drop predictably.

3. The Big Surprise: "Large models overfitting"

In traditional statistics and classical machine learning, there is a fundamental rule: if your model has too many parameters, it will "overfit."

- *Overfitting* means the model gets lazy, memorizes the training data word-for-word like a cheat sheet, and completely fails when asked a new question in the real world.
- The shocker for LLMs: When researchers scaled transformers up to billions of parameters, the models did not overfit. Instead, they actually got better at reasoning, generalization, and understanding abstract concepts.

Before 2020, companies built small, experimental models.

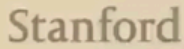
Once this paper proved that scaling up compute, data, and parameters yields guaranteed returns without hitting an immediate wall, the entire industry shifted. It gave companies like OpenAI, Google, and Meta the scientific confidence to invest billions of dollars building massive clusters of GPUs to train models on trillions of tokens directly leading to GPT-4, LLaMA, and modern generative AI.

1. Scaling Laws: Tuning

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Scaling laws: tuning

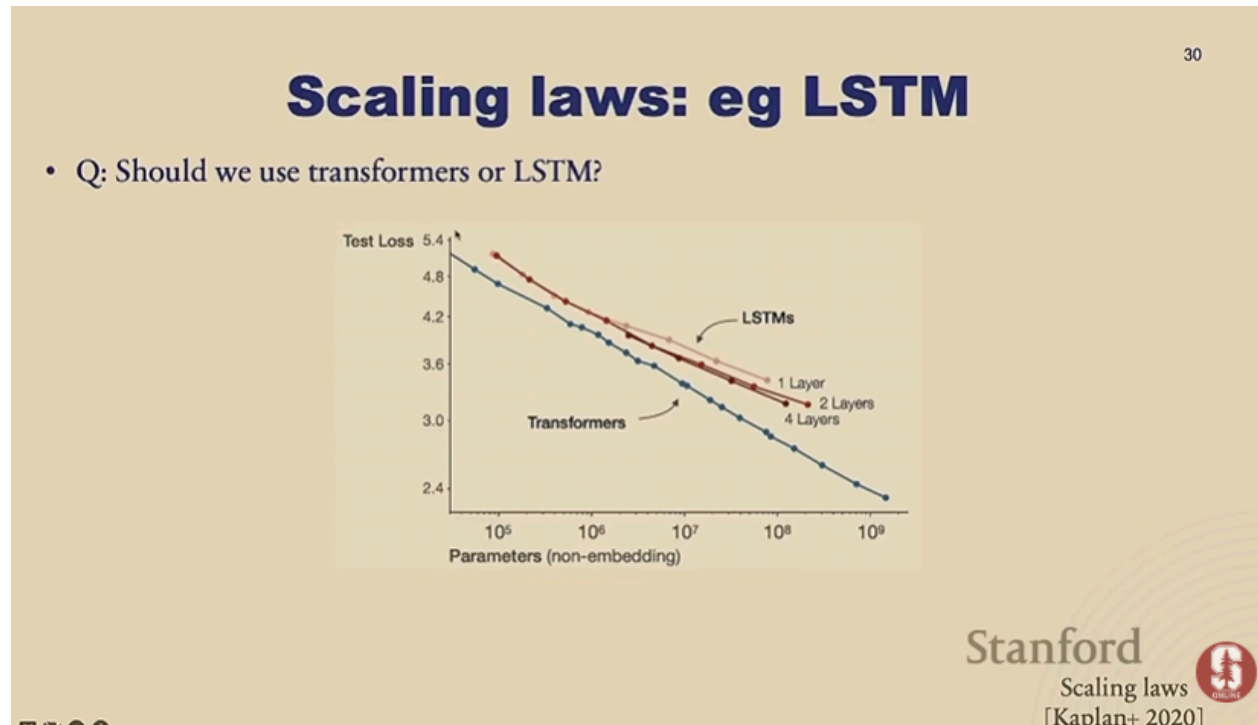
- You have 10K GPUs for a month, what model do you train?
- Old pipeline:
 - Tune hyperparameters on big models (e.g. 30 models)
 - Pick the best => final model is trained for as much as each filtered out ones (e.g. 1 day)
- New pipeline:
 - Find scaling recipes (eg lr decrease with size)
 - Tune hyperparameters on small models of different sizes (e.g. for <3 days)
 - Extrapolate using scaling laws to larger ones
 - Train the final huge model (e.g. >27 days)



If a company has 10,000 GPUs rented for a month, how do they ensure they don't waste that expensive compute time training a bad model?

- The Old Pipeline (Trial and Error): Previously, researchers would train many large models (e.g., 30 different versions), tweaking the settings (hyperparameters) on each one. They would pick the best one, which meant they wasted most of their budget training 29 models that were eventually thrown away.
- The New Pipeline (Using Math): Thanks to scaling laws, researchers now run cheap, fast experiments on very small models (taking less than 3 days). Because they know AI performance scales predictably, they can use math to *extrapolate* exactly how those settings will behave on a massive scale. They then confidently spend the rest of their budget (over 27 days) training one final, massive model, knowing the settings are already optimal.

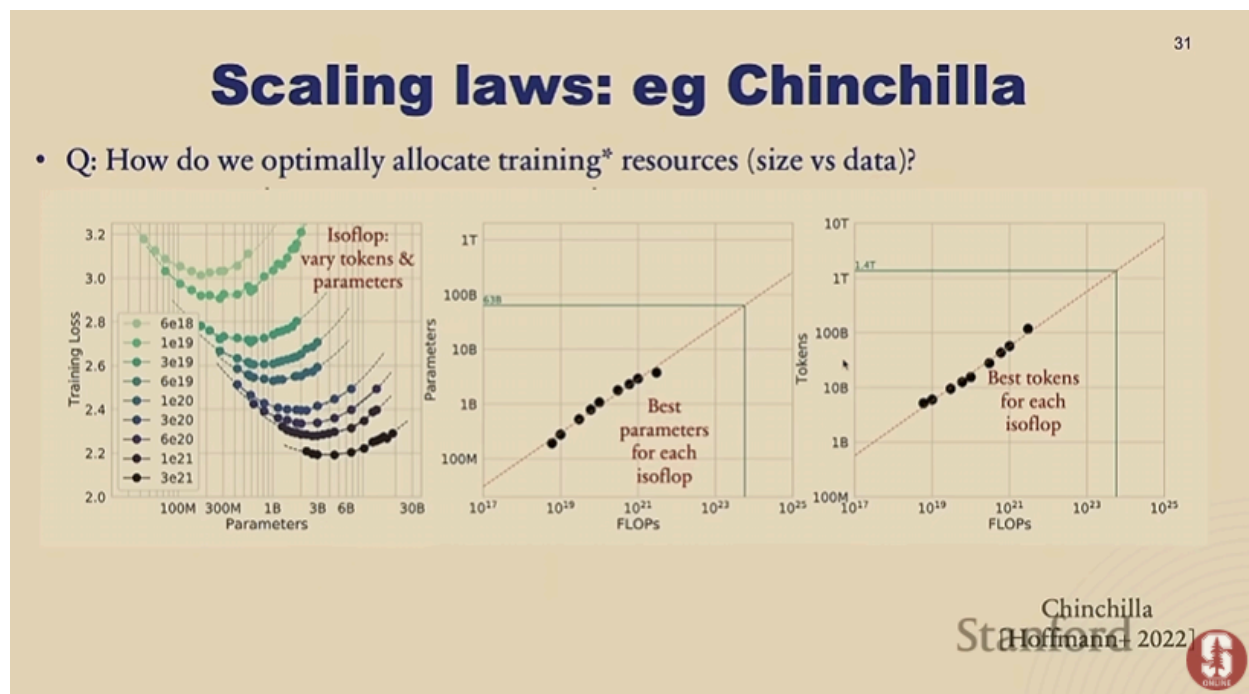
2. Transformers vs. LSTMs



- LSTMs were an older type of AI used for language.
- Transformers are the architecture behind modern models like GPT.

The graph proves why Transformers won. As both models get bigger (moving right on the X-axis), the error rate (Test Loss) drops for both. However, the Transformer line drops steeper and faster than the LSTM lines. Scaling laws proved that Transformers make better use of increased parameters than older technologies.

3. The "Chinchilla" Scaling Laws



A famous 2022 paper from DeepMind called the "Chinchilla" paper. It answered the ultimate question: *If you have a fixed budget of compute (FLOPs), how should you balance the size of the model (parameters) versus the amount of data (tokens)?*

- Before Chinchilla, researchers were building massive models with billions of parameters but not feeding them enough data.
- The Chinchilla paper proved mathematically that model size and dataset size must scale equally.
- If you double the size of your model (parameters), you must also double the amount of reading material (tokens) you feed it to get optimal performance. This realization drastically changed how modern open-source and commercial models are trained today.

Bitter lesson

- **Bitter lesson:** models improve with scale & Moore's Law

=> "only thing that matters in the long run is the leveraging of computation."

Bitter [Sutton 2019] <http://www.incompleteideas.net/Incldeas/BitterLesson.html>

- Don't spend time over complicating: do the simple things and scale them!

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Example : SOTA (State of the Art) model

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Training a SOTA model

- Example of current SOTA: LLaMA 3 400B ~40 tok/param => train compute optimal

Data: 15.6T tokensParameters: 405B
- FLOPs: $6NP = 6 * 15.6e12 * 405e9 = 3.8 e25$ FLOPs ~2x less than executive order
- Compute: 16K H100 with average throughput of 400 TFLOPS
- Time: $3.8e25 / (400e12 * 3600) = 26M$ GPU hour / $(16e3 * 24) = 70$ days From paper: ~30M
- Cost: rented compute + salary = $\sim \$2/h * 26Mh + 500k/y * 50 \text{ employees} = \$52M + \$25M = \sim \$75M$ \$65-85M
- Carbon emitted = $26Mh * 0.7kW * 0.24kg/kWh = 4400$ tCO₂eq ~2k return tickets JFK-LHR
- Next model? ~10x more FLOPs

1. The Model & The Data

- SOTA: Stands for State of the Art (the cutting edge).
- Parameters (P): 405 Billion parameters.
- Dataset (N): 15.6 Trillion tokens.
- The Ratio: 15.6T/405B approx 38.5 (roughly 40 tokens per parameter). Unlike the 8B model which was heavily over-trained for fast consumer inference, Meta trained this 405B giant closer to the compute-optimal balance.

2. The Total Math Work: FLOPs = 6NP

When calculating the raw compute needed to train a transformer, researchers use a standard rule of thumb: $6 * N * P$.

- Why 6? For every single token read, passing forward through the network takes about \$2\$ FLOPs per parameter, and the backward pass (updating weights) takes about \$4\$ FLOPs per parameter. $\$2 + \$4 = \$6$ operations per token per parameter.
- The Total:
That is 38 septillion individual decimal calculations.

- "~2x less than executive order": The US government issued an Executive Order requiring safety reporting for any AI trained above 10^{26} FLOPs. LLaMA 3 405B sat just under that threshold.

3. The Hardware & Time

- Hardware: Meta gathered an array of 16,000 Nvidia H100 GPUs working together in high-speed network clusters.
- GPU Hours: It required roughly 26 million to 30 million GPU hours of total compute time.
- Wall-Clock Time: Even with 16,000 supercomputer chips running in parallel 24 hours a day, the training run took roughly 70 days (over 2 months straight) without stopping.

4. The Financial Bill: ~\$75 Million to \$85 Million

Training a flagship AI is not just a computer science project; it is a massive capital expenditure:

- GPU Rental: Running 26 million GPU hours at roughly \$2/hour costs \$52 Million.
- Engineering Talent: 50 world-class AI research scientists and engineers earning roughly \$500,000/year adds another \$25 Million.
- Total Cost: Somewhere between \$65M and \$85M for a single training run. If an engineering bug or network crash corrupts the run halfway through, millions of dollars vanish.

5. The Environmental Footprint

- Running that many enterprise chips draws immense electricity: 26M GPU hours X 0.7 kW power draw X standard grid emission factors equals roughly 4,400 tonnes of CO2 equivalent.
- That is roughly the carbon footprint of 2,000 round-trip passenger flights between New York (JFK) and London (LHR).

6. "Next model? ~10x more FLOPs"

Every successive generation of frontier models requires roughly 10 times more compute to produce noticeable jumps in capability. This means future frontier models will scale into hundreds of thousands of interconnected GPUs and individual training budgets exceeding hundreds of millions of dollars.

References :

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YouTube.<https://www.youtube.com/watch?v=9vM4p9NN0Ts>
- Hendrycks et al. (2020): Introduced the Massive Multitask Language Understanding (MMLU) benchmark to test models across various academic and professional subjects.
- Hoffmann et al. (2022): Often referred to as the "Chinchilla" paper, this research demonstrated how to optimally allocate training compute by balancing model size and dataset size.
- Kaplan et al. (2020): Established the foundational "Scaling laws," proving that an AI model's performance reliably improves and can be predicted based on the amount of data and the number of parameters.
- Liang et al. (2022): Introduced HELM-lite (Holistic Evaluation of Language Models), an aggregated framework for standardizing NLP benchmarks.
- Sutton (2019): Authored the "Bitter lesson" essay, arguing that leveraging massive computation at scale is the most reliable way to improve AI performance over the long term.